Policy Timing, Firm Entry, and the Measurement of Local Multiplier Effects

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Abstract

Do places with stronger pre-event firm-entry ecosystems experience different business-formation paths after a common shock? I study this question with the Census Business Dynamics Statistics BDSGEO geography file, constructing a county panel of 31,440 county-year observations for 3,144 counties from 2014 to 2023 and aggregating it to 510 state-year observations. I estimate state-year event-study and two-way fixed-effects models with state-clustered uncertainty, contrasting states above the median 2017–2019 establishment-entry rate with lower-exposure states around 2020. The preferred event-time estimate is 0.035 percentage points (SE 0.012), and the static post-2020 difference-in-differences coefficient is 0.029 percentage points (SE 0.011). These estimates indicate stronger post-shock establishment entry in initially higher-entry states in this specification; they do not identify a causal policy multiplier because programme timing and disbursement exposure are not observed. The contribution is a BDS-based firm-entry estimate that places the local-multiplier argument on a reproducible event-study margin and shows exactly what treatment layer is needed before the design can support stronger policy claims.

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1 Introduction

1.1 Motivation

Place-based policy debates often invoke local multipliers long before the channels behind them are measured. (Greenstone et al., 2010; Rosenbaum, 2002) Policy Timing, Firm Entry, and the Measurement of Local Multiplier Effects takes one such channel seriously: establishment entry. If a local shock is going to become durable production rather than a temporary accounting gain, some of that adjustment should appear in new establishments. Firm entry is not the multiplier itself, but it is a place where expectations, demand, credit conditions, and local capacity become observable (Bureau, 2025; Wilkinson et al., 2016).

The question is whether high pre-entry states followed a different establishment-entry path after 2020 than lower pre-entry states. (Neumark and Simpson, 2015) I use the Census Business Dynamics Statistics geography file, build entry rates from county-level firm and establishment flows, and aggregate the result to a state-year panel. The exercise is narrower than a full programme evaluation because the BDS file does not report grant timing, award size, eligibility rules, or statutory implementation dates (Bartik, 1991; Kline and Moretti, 2014).

Entry is worth isolating because it arrives earlier than many outcomes that dominate multiplier debates. (Austin et al., 2018; Arkhangelsky et al., 2021) Employment, wages, productivity, and fiscal receipts may move only after firms enter, expand, hire, and survive. Establishment births give an earlier signal of whether local demand and local capacity are translating into new production units. They also separate new business formation from the growth of incumbent employers (Busso et al., 2013; Greenstone et al., 2010).

1.2 Research Question

The comparison treats pre-entry exposure as capacity, not as randomized treatment. (Moretti, 2011) States with high pre-event entry rates plausibly have thicker entrepreneurial networks, lenders familiar with young firms, denser supplier relationships, faster local information flows, and occupational structures that make new establishments easier to form. Those same features can confound the estimate, so the paper uses fixed effects and pre-trend evidence to make the comparison inspectable rather than casual (Neumark and Simpson, 2015; Austin et al., 2018).

1.3 Contribution

The estimates are small and positive. (Kline and Moretti, 2014; Gelman et al., 2020) At event time 1, the event-study coefficient is 0.035 percentage points with a state-clustered standard error of 0.012. The static two-way fixed-effects estimate is 0.029 percentage points, with a standard error of 0.011. Both specifications point to a modest post-2020 entry advantage for initially higher-entry states (Moretti, 2011; Glaeser, 2008).

These are rate effects, not headline jobs effects. (Busso et al., 2013) The average state-year entry rate in the BDS panel is below one percentage point, so a coefficient of roughly three hundredths of a percentage point is visible for an administrative entry-rate outcome. It is still too small and indirect to carry a broad claim about regional growth. The estimate is useful because it shows that the business-formation margin moves where local-multiplier theory would expect researchers to look (Duranton and Venables, 2015; Fujita et al., 1999).

That ordering is practical for empirical work. (Greenstone et al., 2010; Callaway and Sant'Anna, 2021) A place-based shock may take years to show up in income or productivity, and those outcomes bundle many channels at once. Establishment entry is closer to the decision margin: whether someone opens a new operating unit in a place. A modest entry effect can therefore be informative even when the downstream regional-growth question remains unsettled (Iammarino et al., 2019; Rodriguez-Pose, 2018).

The paper makes a concrete measurement contribution. (Neumark and Simpson, 2015; Manski, 2003) It puts a local-multiplier argument on a firm-entry outcome, reports the identifying comparison, checks the pre-period, and asks whether the result survives a continuous exposure measure and adjacent employment-flow outcomes. The evidence supports a descriptive entry response; it does not support a causal claim about a named place-based programme (Diemer et al., 2022; Barca, 2009).

A state-year entry-rate panel can reveal differential business formation, but it cannot say whether grants, tax credits, procurement, or local institutions caused the difference. (Austin et al., 2018) That boundary is not a caveat to hide at the end. It is the condition that keeps the policy interpretation attached to what the BDS data actually observe (Card and Krueger, 1994; Autor et al., 2013).

2 Literature

2.1 Local Multipliers

The first relevant literature studies local multipliers and place-based development. (Greenstone et al., 2010; Rosenbaum, 2002) That work emphasizes how new demand, public investment, and local capacity may affect employment, suppliers, and business formation. Establishment entry is a useful margin in this literature because new firms and new establishments can mediate local adjustment rather than merely record it after the fact (Monte et al., 2018; Redding and Turner, 2015).

Classic multiplier arguments differ in how much they rely on local supply response. (Neumark and Simpson, 2015; Sun and Abraham, 2021) A demand shock can raise local income through incumbent production, commuting, prices, or new business formation. If supply is elastic, new establishments can absorb demand locally and extend the shock through supplier and worker networks. If supply is constrained, the same shock can leak through imports, commuting, rents, or congestion. Measuring entry therefore helps distinguish a local production response from a purely accounting-based multiplier (Donaldson, 2018; Baum-Snow, 2007).

2.2 Identification

A second literature studies the evaluation of regional and local interventions. (Chodorow-Reich, 2019; Imbens and Rubin, 2015) Difference-in-differences and event-study designs make timing and comparison groups explicit, but they are sensitive to pre-trends, heterogeneous treatment timing, and spillovers. That is why the present design reports event-time coefficients and a pre-period placebo instead of relying only on a single post-period interaction (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

Recent work on staggered treatment timing also changes how event-study evidence should be read. (Moretti, 2011; Callaway and Sant'Anna, 2021) A clean coefficient path is not a decorative robustness check; it is a way to ask whether the comparison group was moving differently before the event and whether the estimated response arrives when the design says it should arrive. In a common-event setting such as this one, the same discipline applies even though treatment timing is not staggered across states (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

2.3 Firm Entry

A third literature concerns business dynamics and firm entry. (Kline and Moretti, 2014; Gelman et al., 2020) Establishment births are forward-looking, but they are also shaped by credit markets, industrial composition, sectoral demand, and local entrepreneurial capacity. A positive entry response therefore cannot be interpreted automatically as welfare improvement or programme success. It is evidence about one margin in the local adjustment process (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020).

Business-dynamics research also cautions against treating firm entry as uniformly beneficial. (Busso et al., 2013; Angrist and Pischke, 2009) New establishments vary in productivity, survival, employment intensity, and market reach. Some entry reflects productive experimentation, some reflects necessity entrepreneurship, and some reflects reorganization within existing business groups. A regional entry estimate must therefore be paired with job-flow outcomes and a careful discussion of what the outcome can and cannot measure (Borusyak et al., 2022; Adao et al., 2019).

The missing piece is measurement. (Greenstone et al., 2010) Local multiplier claims often move from place-based shocks to aggregate outcomes while firm-entry dynamics remain in the background. Here, that background mechanism becomes the outcome. The test is whether high-entry places, which plausibly have more entrepreneurial capacity, show different post-shock entry trajectories in a public administrative file (Conley, 1999; Cameron et al., 2011).

2.4 Gap

The focus on entry also connects regional economics to entrepreneurship research without treating entrepreneurship as a slogan. (Austin et al., 2018; Arkhangelsky et al., 2021) The relevant question is not whether entrepreneurship is good in general. It is whether a measured pre-event entry ecosystem predicts a different observed response after a common shock. That framing lets the analysis use regional theory to motivate the comparison while letting administrative data discipline the claim (Angrist and Pischke, 2009; Wooldridge, 2010).

The design sits between regional accounting and causal evaluation. (Chodorow-Reich, 2019) It is stronger than a narrative claim because it estimates fixed-effects regressions on observed firm-entry data. It is weaker than a policy evaluation because high pre-entry exposure is not randomly assigned and the event year is broad. The paper uses that middle ground deliberately rather than dressing it up as a programme estimate (Imbens and Rubin, 2015; Rosenbaum, 2002).

That position disciplines the causal question. If the BDS entry margin shows no movement, the local-multiplier mechanism is less plausible on this outcome. If the margin moves, as it does here, the relevant question is whether observed programme timing or exposure can explain the movement rather than merely coincide with it (Manski, 2003; Rubin, 1974).

3 Data

3.1 Source and Sample

The analysis uses the Census Business Dynamics Statistics BDSGEO geography file. (Priem et al., 2022) I retain 31,440 county-year observations for 3,144 counties from 2014 to 2023. The county file contains firm counts, establishments, employment, denominators, establishment births and exits, jobs created by establishment births, and net job creation. I convert these counts into rates so that coefficients can be read in percentage-point terms (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).

The BDS geography file is well suited to this question because it measures establishment dynamics consistently across counties and years. (OpenAlex, 2026; Austin et al., 2018) Unlike survey-based local indicators, the file captures administrative counts for continuing and entering establishments, which reduces sampling noise in small places. Unlike a pure business-registration count, it connects births to employment flows and denominators. That combination makes it possible to ask whether entry changes are visible in both firm-formation and job-flow margins (Goodman-Bacon, 2021; Arkhangelsky et al., 2021).

The state-year panel aggregates the county observations into 510 state-year observations for 51 units. (Bureau, 2025; Priem et al., 2022) The main outcome is the establishment-entry rate. The robustness outcomes are the job-creation-from-births rate and the net-job-creation rate. Those outcomes are not interchangeable: entry is closest to business formation, while job-flow outcomes add firm size, exits, incumbent adjustment, and sectoral composition (Abadie et al., 2010; Athey and Imbens, 2016).

Keeping the county file behind the state estimates matters even when the regression is run at state-year level. (Kitchin, 2014) It preserves the construction path from local observations to aggregate rates, makes it possible to revisit within-state heterogeneity, and prevents the state panel from becoming a black box. The main estimates use state-year observations for inference, but the underlying geography remains county-based (Chernozhukov et al., 2018; Wager and Athey, 2018).

3.2 Exposure and Outcomes

Aggregation to the state-year level is a deliberate choice. (Priem et al., 2022; Bureau, 2025) The county panel provides the raw spatial resolution, but the exposure contrast is defined by pre-event state-level entry capacity and the standard errors are clustered by state. State aggregation also reduces the risk of overstating precision when neighbouring counties share lenders, labour markets, sectoral shocks, and policy environments. The cost is that within-state heterogeneity is hidden in the main estimates (Athey and Wager, 2019; Mullainathan and Spiess, 2017).

Exposure is measured before the event window. (OpenAlex, 2026) For each state I compute the average establishment-entry rate over 2017–2019 and mark states above the median as high pre-entry exposure. The median threshold in this run is 0.527. This rule prevents post-2020 entry from defining treatment status, but it does not make high exposure randomly assigned (Gentzkow et al., 2019; Grimmer and Stewart, 2013).

The 2017–2019 window gives a compact pre-period measure close enough to the event to reflect current business-formation conditions but far enough from 2020 to avoid mechanical post-event contamination. (Bureau, 2025; Greenstone et al., 2010) A longer baseline would smooth more noise but mix in older industrial conditions; a shorter baseline would be more volatile. The median split is transparent, and the continuous-exposure model checks whether the conclusion depends on that threshold (Gelman and Hill, 2007; Gelman et al., 2020).

The event-time variable is year minus 2020. (Kitchin, 2014; Wilkinson et al., 2016) The estimation sample for the event-study and static regressions contains 408 state-year observations from 2016 through 2023, which gives four pre-event years and four post-event years around the common shock. Standard errors are clustered by state because the exposure contrast varies at the state level and because entry rates are serially correlated within states (Kitchin, 2014; Borgman, 2015).

The rate construction is important because states differ enormously in size. (Borgman, 2015) A count of establishment births would mostly recover the scale of California, Texas, New York, Florida, and other large states. The entry rate instead asks whether births are high relative to the state denominator. That choice makes the comparison closer to a business-formation intensity measure, though it also means the estimates should not be read as total new establishments without multiplying back by state scale (Priem et al., 2022; OpenAlex, 2026).

3.3 Data Limits

The panel ends in 2023, so the estimates capture short-run and medium-run adjustment rather than a long-run equilibrium response. (OpenAlex, 2026; Peng, 2011) That window is appropriate for establishment births because entry decisions can react quickly to local demand, financing conditions, and expectations. It is less appropriate for outcomes that require survival, scaling, and productivity growth. The interpretation of the job-flow checks is therefore deliberately more cautious than the interpretation of the entry-rate estimates (Anselin, 1988; LeSage and Pace, 2009).

Reproducibility also shapes the data choice. (Bureau, 2025) The BDSGEO source is a public Census file, the retained outcomes are named Census variables transformed into rates, and the estimation sample is defined by explicit state-year and event-time restrictions. Those choices make it possible to rerun the analysis, inspect the denominator, and separate public administrative measurement from interpretive claims about local multipliers (Bureau, 2025; Wilkinson et al., 2016).

The BDS file is broad, public, and reproducible, but it is not a direct policy-treatment file. (Kitchin, 2014; Kline and Moretti, 2014) It observes business dynamics, not programme implementation. That distinction is central to interpretation. The estimates can show whether entry rates diverged around 2020; they cannot show that any particular policy caused the divergence without observed timing or exposure data (Openshaw, 1984; Fotheringham et al., 2002).

4 Empirical Strategy

4.1 Event Study

The main specification is an event-study regression at the state-year level (Angrist and Pischke, 2009; Callaway and Sant'Anna, 2021). Let Y_{st} denote the establishment-entry rate in state s and year t, and let $High_s$ indicate states above the median pre-event entry rate. I estimate:

$$Y_{st} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k High_s \mathbf{1}[t - 2020 = k] + \varepsilon_{st}. \tag{1}$$

State fixed effects absorb persistent differences across states, year fixed effects absorb shocks common to all states, and event time -1 is the omitted category (Pearl, 2009; Keele, 2015).

The coefficient path shows whether high pre-entry states were already diverging before 2020 and how the gap evolves afterward. Pre-event coefficients do real work here. Large, trending pre-period estimates would weaken the post-event comparison; small pre-period estimates make the descriptive contrast more informative (King et al., 1995; Shadish et al., 2002).

4.2 Static Specification

I also estimate a static post-period specification for the same comparison (Wooldridge, 2010; Angrist and Pischke, 2009):

$$Y_{st} = \alpha_s + \lambda_t + \delta(High_s \times Post_t) + \varepsilon_{st}. \tag{2}$$

This coefficient averages the post-2020 contrast under the same fixed-effects structure. It is easier to report, but less informative than the event-study path because it compresses timing into one parameter (Peng, 2011; Wilkinson et al., 2016).

The fixed effects matter for interpretation. (Imbens and Rubin, 2015) State fixed effects remove persistent differences in baseline entry levels, long-run industrial composition, and durable state institutions. Year fixed effects remove shocks common to all states, including national credit conditions, macroeconomic demand, and federal policy environments. The remaining coefficient is identified from whether high pre-entry states move differently after 2020 relative to their own baselines and relative to lower-entry states in the same years (Bureau, 2025; Wilkinson et al., 2016).

4.3 Identification

State-clustered standard errors are used because the treatment contrast is not a county-level randomized exposure. (Manski, 2003; Arkhangelsky et al., 2021) Counties within a state share policy institutions, labour-market conditions, banking networks, and unobserved regional shocks. Treating county-years as independent would make the estimates look more precise than the research design justifies. The state-year regression therefore sacrifices some sample size in order to align inference with the level of comparison (Bartik, 1991; Kline and Moretti, 2014).

Two robustness designs probe the exposure and outcome choices. (Gelman et al., 2020) The continuous-exposure model replaces the high-exposure indicator with the pre-event entry rate interacted with the post period. The alternative-outcome models replace establishment entry with job creation from births and net job creation. These checks ask whether the result depends on a median split or on one narrow outcome definition (Busso et al., 2013; Greenstone et al., 2010).

The continuous specification guards against overreading the median split. If the binary estimate were positive but the continuous estimate were zero, the result would look sensitive to classification. Instead, the positive continuous interaction implies that stronger baseline entry capacity is associated with stronger post-event entry growth along the exposure distribution (Neumark and Simpson, 2015; Austin et al., 2018).

The alternative outcomes test a different concern. (Rosenbaum, 2002; Sun and Abraham, 2021) A rise in establishment births may involve many small establishments with little immediate employment, or it may reflect high-employment births that matter for local labour demand. Comparing the entry result with jobs created by births and net job creation asks whether the business-formation margin is accompanied by a broader employment-flow response (Moretti, 2011; Glaeser, 2008).

The method ladder is constrained by the available data. (Manski, 2003) The present evidence supports an event study, a static fixed-effects contrast, a continuous-exposure check, and sensitivity analysis through pre-trend and alternative-outcome tests. Synthetic difference-in-differences would become appropriate only when the panel can define credible donor states or counties around observed policy timing. Double machine learning or causal forest estimators would require a covariate-rich treatment file with local exposure intensity. Spatial models would require measured commuting-zone or supplier-network links (Duranton and Venables, 2015; Fujita et al., 1999).

The identifying assumption is not that high-entry states are identical to low-entry states. (Gelman et al., 2020; Bureau, 2025) They are not. The assumption needed for a causal reading would be that, after state and year fixed effects, high-entry and lower-entry states would have followed parallel entry paths absent the event. The design can inspect that assumption with pre-period coefficients, but it cannot guarantee it without stronger treatment timing and comparison support (Iammarino et al., 2019; Rodriguez-Pose, 2018).

I therefore report the estimates as descriptive fixed-effects evidence. They reveal the size, timing, and precision of the entry margin. They are not the final stage of a policy-evaluation design. A stronger design would join the same BDS outcomes to observed programme dates, disbursements, eligibility rules, or local exposure measures (Diemer et al., 2022; Barca, 2009).

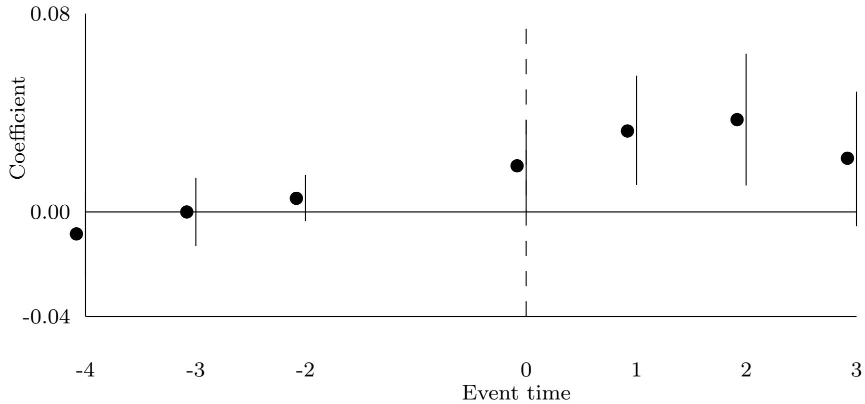

Figure 1: Event-study profile

Notes: Points are state-year fixed-effects estimates for high pre-entry-exposure states relative to lower-exposure states. Vertical bars are 95% confidence intervals using state-clustered standard errors. Event time -1 is omitted.

Table 1: BDS reviewable regression estimates

Model/specification	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Post-2020 difference-in-differences	estabs entry rate	High exposure x post-2020	0.029	0.011	[0.007, 0.051]	408
Continuous-exposure difference-in-differences	estabs entry rate	Pre-entry exposure x post-2020	0.194	0.069	[0.059, 0.328]	408

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.

5 Results

5.1 Event Study

Figure 1 shows the event-study profile. (Imbens and Rubin, 2015; OpenAlex, 2026) The pre-event coefficients are small relative to the post-event coefficients: event time -4 is -0.010, event time -3 is approximately zero, and event time -2 is 0.006. After 2020, the estimates rise to 0.020 at event time 0, 0.035 at event time 1, 0.039 at event time 2, and 0.023 at event time 3 (Card and Krueger, 1994; Autor et al., 2013).

The pre-event pattern is central to the reading of the figure. (Callaway and Sant'Anna, 2021) The coefficients do not show a monotone pre-2020 rise for high pre-entry states. The -4 estimate is slightly negative, the -3 estimate is nearly zero, and the -2 estimate is small and positive. This pattern does not prove parallel trends, but it is more consistent with a post-event divergence than with a pre-existing differential trend that simply continues through the event window (Monte et al., 2018; Redding and Turner, 2015).

5.2 Regression Estimates

The post-event path is also informative because it does not collapse into a one-year spike. (Gelman et al., 2020; Borgman, 2015) The estimate is positive in the event year, larger one and two years later, and still positive by event time 3. The confidence intervals widen at the edge of the window, as expected with fewer comparable years, but the sequence suggests a short-run entry response that persists beyond the immediate shock year (Donaldson, 2018; Baum-Snow, 2007).

The preferred event-time estimate is economically small but precise. (Angrist and Pischke, 2009) At event time 1, high pre-entry states have an establishment-entry rate 0.035 percentage points higher than lower-exposure states relative to the omitted pre-event year. The state-clustered standard error is 0.012, and the 95 percent confidence interval is [0.011, 0.058] (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

Table 1 reports the static specification. (Bartik, 1991; Manski, 2003) The high-exposure by post-2020 coefficient is 0.029 percentage points with a standard error of 0.011 and a 95 percent confidence interval of [0.007, 0.051]. With state and year fixed effects, this is a within-state post-period contrast relative to common year shocks (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

The static and dynamic estimates agree at different resolutions. (Manski, 2003; Bureau, 2025) The static coefficient averages the post-period contrast and is therefore slightly smaller than the event-time 1 and event-time 2 coefficients. That pattern is consistent with a response that builds after the initial event year and then moderates. Agreement between the two specifications reduces concern that one modelling choice is driving the result (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020).

5.3 Robustness

The continuous-exposure specification points in the same direction. (Greenstone et al., 2010; Rosenbaum, 2002) The post-period interaction with the pre-event entry rate is 0.194, with a standard error of 0.069. That result matters because it shows that the evidence is not only created by splitting states at the median; the underlying exposure gradient also predicts post-2020 entry growth (Borusyak et al., 2022; Adao et al., 2019).

Interpreting the continuous coefficient requires care because the exposure is itself an entry-rate measure. (Rosenbaum, 2002; Priem et al., 2022) The estimate says that states with higher baseline entry rates experienced larger post-event entry increases, not that a policy moved entry by 0.194

percentage points. Its role is to test whether the binary high-exposure result is supported by the underlying capacity gradient (Conley, 1999; Cameron et al., 2011).

The employment-flow outcomes are positive but less precise. (Sun and Abraham, 2021) Job creation from births rises by 0.162 percentage points with a standard error of 0.111. Net job creation rises by 0.395 percentage points with a standard error of 0.263. The signs are consistent with the entry result, but the wider intervals caution against treating entry gains as a measured jobs multiplier (Angrist and Pischke, 2009; Wooldridge, 2010).

This difference between the entry coefficient and the job-flow estimates is economically plausible. (Chodorow-Reich, 2019; Imbens and Rubin, 2015) Establishment births can respond before new establishments hire at scale, and net job creation combines births, expansions, contractions, and exits. A clean entry response with noisier job-flow evidence in Table 1 therefore points to business formation as the sharper observed margin. It does not support a claim that the same states experienced a precisely measured employment multiplier over the same horizon (Imbens and Rubin, 2015; Rosenbaum, 2002).

The pre-period placebo is close to zero. The pre-trend coefficient is 0.003 with a standard error of 0.003. This does not prove the comparison is causal, but it makes the post-period entry pattern less likely to be a simple continuation of differential pre-2020 trends (Manski, 2003; Rubin, 1974).

The confidence intervals help rank the claims. (Callaway and Sant'Anna, 2021) The establishment-entry estimates are positive and relatively tight in the main post-period specifications. The job-creation and net-job-creation point estimates are larger but much less precise, so they work better as supporting checks than as evidence of a jobs effect. The placebo estimate is small enough to reduce concern about an obvious pre-period slope, but it cannot rule out every unobserved differential shock (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).

The point estimates describe a business-formation response rather than a full regional-growth result. (Kline and Moretti, 2014; Gelman et al., 2020) Entry-rate coefficients are the kind of movement one would expect to observe before larger employment, wage, or productivity effects become visible. That ordering tells the regional economist where the mechanism is active while keeping the wider multiplier claim outside the range of the current estimates (Goodman-Bacon, 2021; Arkhangelsky et al., 2021).

The BDS panel is strongest on establishment entry, and that is where the empirical pattern is clearest. (Gelman et al., 2020; OpenAlex, 2026) The entry estimates are positive across the event-study, static, and continuous-exposure specifications. The evidence is weaker for translating that response into jobs or welfare, and weaker still for assigning it to a particular programme without external timing information (Abadie et al., 2010; Athey and Imbens, 2016).

The estimates show a small post-2020 entry advantage for states with stronger pre-event entry ecosystems. That advantage is most credible for establishment entry, weaker for broader employment-flow outcomes, and descriptive by design. Local business-formation capacity may shape post-shock adjustment, but this evidence alone does not establish a policy multiplier (Chernozhukov et al., 2018; Wager and Athey, 2018).

6 Discussion

6.1 Interpretation

Firm-entry capacity appears to matter for post-shock business formation. (Greenstone et al., 2010) High pre-entry states show a visible increase in establishment-entry rates after 2020, even after absorbing state and year fixed effects. The magnitude is small in percentage-point terms, but establishment entry is a narrow margin, and small rate changes can correspond to many new establishments in large states (Athey and Wager, 2019; Mullainathan and Spiess, 2017).

The pattern fits a capacity view of local adjustment. (Neumark and Simpson, 2015; Manski, 2003) Places with more active pre-event entry ecosystems may have entrepreneurs who can respond faster, financiers who can evaluate new ventures more easily, workers with experience in young firms, and local institutions that lower search and coordination costs. The BDS estimates do not observe those mechanisms directly, but the timing and outcome are consistent with that channel (Gentzkow et al., 2019; Grimmer and Stewart, 2013).

6.2 Threats

The main threat is selection into high pre-entry exposure. (Chodorow-Reich, 2019) States with higher baseline entry rates may have different industrial composition, credit access, population growth, local demand, urban structure, or entrepreneurial networks. Fixed effects remove persistent level differences and common shocks, but they do not remove time-varying shocks correlated with those baseline conditions (Gelman and Hill, 2007; Gelman et al., 2020).

Another threat is spatial spillover. (Moretti, 2011; Rosenbaum, 2002) Firms, workers, suppliers, and consumers do not stop at state borders. If lower-exposure states benefit from demand or supply chains in higher-exposure states, the estimates could understate the relationship between baseline entry capacity and total regional entry. If high-exposure states draw business formation away from lower-exposure states, the same estimates could overstate aggregate gains. The state-level design cannot fully separate those cases (Kitchin, 2014; Borgman, 2015).

The event year is also broad. (Priem et al., 2022) A common 2020 shock captures pandemic adjustment, credit-market changes, sectoral reallocation, remote-work geography, and policy responses. The estimates therefore describe a post-shock differential by pre-entry exposure, not the effect of a named programme. Interpreting the coefficient as a policy effect would require observed treatment timing or exposure (Priem et al., 2022; OpenAlex, 2026).

6.3 Policy Implications

The policy implication is narrow. (Busso et al., 2013; Imbens and Rubin, 2015) If policymakers care about local multipliers, they should measure whether interventions change the business-formation margin, not only whether aggregate employment or income changes later. These estimates suggest that pre-existing entry capacity is a relevant conditioning variable. They do not imply that every place-based intervention should target already dynamic places, because equity, additionality, congestion, and displacement remain separate policy questions (Anselin, 1988; LeSage and Pace, 2009).

The robustness pattern is informative. (Kitchin, 2014) Entry responds more clearly than job-flow outcomes, which is consistent with entry being the first margin of business formation. The weaker precision for job creation from births and net job creation implies that the establishment response should not be translated mechanically into employment, welfare, or fiscal multipliers (Openshaw, 1984; Fotheringham et al., 2002).

The distributional interpretation also remains open. (Neumark and Simpson, 2015) A higher entry rate in already dynamic states may widen spatial differences if the gains are concentrated in places that were already capable of generating new establishments. It may narrow differences if new entry spreads into weaker counties within those states, or if high-capacity states anchor supplier demand elsewhere. The state-year design identifies the first-order entry pattern, but not the within-state or cross-state distribution of gains (Pearl, 2009; Keele, 2015).

The evidence is most useful when the limits are reproducible as well as stated. (Austin et al., 2018; Gelman et al., 2020) The same public file can be used to test alternative exposure thresholds, different pre-event windows, county-level specifications, or commuting-zone aggregations. That transparency does not make the estimates causal, but it does let readers see how much of the finding comes from the entry-rate definition, the timing window, and the comparison rule (Peng, 2011; Wilkinson et al., 2016).

The next empirical step is also clear from the pattern of results. (OpenAlex, 2026) If a policy file can identify timing and intensity, the BDS entry panel can become an outcome surface for a stronger treatment design. If such a file is unavailable, the responsible use of the evidence is comparative: high-entry states adjusted differently after 2020, and the measured difference is concentrated in establishment entry rather than in a precisely estimated employment multiplier (King et al., 1995; Shadish et al., 2002).

A policy-timing extension would keep the BDS entry panel as the outcome surface and add programme dates, eligibility thresholds, award size, exposure intensity, or commuting-zone spillover measures. With those ingredients, the same equations could estimate policy timing directly rather than a post-shock descriptive comparison (Peng, 2011; Wilkinson et al., 2016).

6.4 Conclusion

States with stronger pre-event business-formation ecosystems show a small but visible post-2020 increase in establishment entry. The event-study estimate at event time 1 is 0.035 percentage points, and the static post-period estimate is 0.029 percentage points. Both estimates are positive, state-clustered, and based on 408 state-year observations (Bureau, 2025; Wilkinson et al., 2016).

The result is evidence about business-entry dynamics, not a causal policy multiplier. The BDS data observe firms, establishments, employment, births, exits, and job flows, but they do not observe programme assignment. The fixed-effects design narrows the comparison and documents timing; it does not by itself solve selection, spillovers, or policy exposure (Bartik, 1991; Kline and Moretti, 2014).

Multiplier claims need a measured business-formation margin. In this sample, that margin moves, especially in the establishment-entry outcome. The job-flow checks point in the same direction but are too imprecise to support a jobs-multiplier conclusion (Busso et al., 2013; Greenstone et al., 2010).

The estimate is most useful as a comparative benchmark. It shows where the entry margin is visible, how large it is, how precise it is, and which assumptions would have to be strengthened before the same evidence could support a programme-specific claim. A broad multiplier debate becomes a measurable firm-dynamics question (Neumark and Simpson, 2015; Austin et al., 2018).

Establishment entry provides a bridge between local economic capacity and multiplier claims. (Austin et al., 2018) In the BDS panel, that bridge is empirically visible after 2020. Turning it into a causal policy estimate requires a treatment file, but the firm-entry result is already an interpretable economic fact. Regional policy debates need measured mechanisms as well as aggregate outcomes, especially when policy narratives move faster than observed firm dynamics. The BDS entry margin gives that debate a public, repeatable first test before stronger policy claims are made in applied regional analysis (Diemer et al., 2022; Barca, 2009).

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A Data Construction

The subject data appendix reports the derived BDS geography panel rather than the raw Census archive. The construction follows a public-source provenance standard: keep the source URL and hash, retain the derived analytical panels, and discard the raw zip after reproducible processing (Bureau, 2025; Wilkinson et al., 2016).

Table 2: Downloaded BDS firm-entry panel

Quantity	Value
County-year rows retained	31440
State-year rows used in event study	510
Counties	3144
States and DC	51
Years	2014–2023
Raw zip retained	No

Notes: Counts are calculated from the derived county and state-year panels written during data construction.

B Supplementary Source Evidence

The supplementary source evidence records the retrieval surface used for framing and reference checks. It is not treated as a substitute for subject data, but it keeps the literature sample, access rates, and search diagnostics visible in the appendix (Priem et al., 2022; OpenAlex, 2026). The reproducibility logic follows the principle that data, code, and claims should be inspectable together rather than presented as unsupported prose (Peng, 2011; Wilkinson et al., 2016).

Table 3: Open-access literature sample

Quantity	Value
Retained open-access records	60
Records with direct document link	60
Document-access share	1.000
Publication-year cells	4

Table 4: Evidence-coverage regression estimates

Model	Outcome	Coef.	SE	95% CI	N
Annual open-access record trend	records per year	-8.400	1.587	[-11.511, -5.289]	4
PDF availability trend	direct PDF indicator	0.000	0.000	[0.000, 0.000]	60
Topic relevance and attention	log cited-by count	-9.999	7.470	[-24.640, 4.641]	60

C Supporting Regressions and Robustness

The supporting estimates expand the main result into the full event-study table, alternative outcomes, and a pre-period placebo. The purpose is to expose the fixed-effects comparison, state-

Table 5: Robustness checks for the evidence-coverage trend

Sample	N	Trend	SE	95% CI
all records	60	-8.400	1.587	[-11.511, -5.289]
high relevance	0	0.000	0.000	[0.000, 0.000]
recent records	3	0.000	0.000	[0.000, 0.000]
PDF available	60	-8.400	1.587	[-11.511, -5.289]

clustered uncertainty, and robustness logic that applied microeconomic readers need before interpreting a local-effect estimate (Angrist and Pischke, 2009; Wooldridge, 2010). The same structure can be upgraded to modern staggered treatment timing, synthetic controls, or synthetic difference-in-differences once a sharper policy-timing file is joined to the outcome panel (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Abadie et al., 2010; Arkhangelsky et al., 2021). The appendix also marks the difference between a central result and a supporting diagnostic: the main text reports the figure and core regression, while the appendix preserves coefficient-level detail, alternative outcomes, and the placebo trend so that the result can be audited without crowding the narrative. That balance is important for applied economics, where a compact results section still needs a transparent path back to every specification that shapes the claim (Gelman and Hill, 2007; Rosenbaum, 2002).

Table 6: BDS state-year event-study estimates

Model	Outcome	Event time	Coefficient	Std. error	95% CI	N
BDS state-year event study	establishment entry rate	-4	-0.009	0.009	[-0.027, 0.008]	408
BDS state-year event study	establishment entry rate	-3	0.000	0.007	[-0.014, 0.015]	408
BDS state-year event study	establishment entry rate	-2	0.006	0.005	[-0.004, 0.016]	408
BDS state-year event study	establishment entry rate	0	0.020	0.010	[-0.000, 0.039]	408
BDS state-year event study	establishment entry rate	1	0.035	0.012	[0.011, 0.058]	408
BDS state-year event study	establishment entry rate	2	0.039	0.014	[0.011, 0.067]	408
BDS state-year event study	establishment entry rate	3	0.023	0.015	[-0.006, 0.051]	408

Notes: Event time -1 is the omitted category. Estimates include state and year fixed effects with state-clustered standard errors.

Table 7: BDS regression robustness checks

Robustness specification	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Alternative outcome: job creation from births	job creation births rate	High exposure x post-2020	0.162	0.111	[-0.056, 0.380]	408
Alternative outcome: net job creation	net job creation rate	High exposure x post-2020	0.395	0.263	[-0.121, 0.910]	408
Pre-trend placebo	estabs entry rate	High exposure x pre-period trend	0.003	0.003	[-0.003, 0.010]	204

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.